

Software Requirements Specification (SRS)

Hospital Management System

Software Development Team

1. Project Overview

The Hospital Management System is a web-based application designed to digitize and automate the daily operations of a hospital. The system aims to improve patient care, reduce paperwork, streamline communication between departments, and maintain accurate medical records.

The application will support multiple user roles, including administrators, doctors, nurses, receptionists, pharmacists, laboratory staff, and patients.

The system should be scalable, secure, and capable of handling multiple hospitals in future versions.

2. Project Objectives

The system should:

Digitize hospital operations.

Maintain electronic medical records. Allow online appointment booking. Manage doctors and hospital staff. Handle billing and payments. Manage laboratory reports. Manage pharmacy inventory.

Generate reports for hospital administration.

3. User Roles

Administrator Responsibilities:

Manage hospital information. Add, edit, and remove doctors. Add and manage staff.

Manage departments.

View reports.

Manage system settings.

Assign user roles.

Doctor Responsibilities:

View appointments.

View patient medical history.

Diagnose patients.

Create prescriptions.

Request laboratory tests.

Update patient records.

Receptionist Responsibilities:

Register patients.

Schedule appointments.

Cancel appointments.

Manage patient information.

Generate bills.

NurseResponsibilities:

View assigned patients.

Record patient vitals.

Update treatment progress.

Assist doctors.

Pharmacist Responsibilities:

Aa medicine inventory.

Dispense medicines.

Update stock.

Generate medicine purchase reports.

Laboratory Technician Responsibilities:

View pending test requests.

Record laboratory results.

Upload laboratory reports.

Patient Responsibilities:

Register account.

Login securely.

Book appointments.

View prescriptions.

View laboratory reports.

Download invoices.

View medical history.

4. Functional Requirements

Patient Management

The system shall:

Register new patients.

Generate unique Patient IDs.

Store patient profile.

Store emergency contact.

Store insurance information.

Maintain medical history.

Search patients by name or ID.

Doctor Management

The system shall:

Add doctors.

Assign departments.

Manage schedules.

View appointments.
Manage availability.

Appointment Management

The system shall:

Book appointments.
Cancel appointments.
Reschedule appointments.
Prevent double booking.
Send appointment confirmations.

Medical Records

The system shall store:

Diagnoses
Allergies
Medications
Surgeries
Treatment history
Vaccination history
Medical notes
Prescription Module

Doctors shall be able to:

Create prescriptions.
Select medicines.
Specify dosage.
Specify duration.
Print prescriptions.
Laboratory Module

Doctors can:

Request laboratory tests.

Laboratory staff can:

Upload reports.
Update test status.
Attach PDF reports.

Patients can:

Download reports.
Pharmacy Module

The system shall:

Store medicine inventory.
Track stock quantity.
Alert low stock.
Manage suppliers.
Record medicine sales.

Billing Module

The system shall:

Generate invoices for:
Consultation fees
Laboratory tests
Medicines
Hospital room charges
Surgery charges
Payment methods:
Cash
Credit Card
Debit Card
Insurance

5. Database Entities

The system should maintain the following entities:

Users
Patients
Doctors
Departments
Appointments
Medical Records
Prescriptions
Medicines
Pharmacy Inventory
Laboratory Tests
Laboratory Reports
Bills
Payments

6. Security Requirements

The system shall:

Use JWT authentication.
Encrypt passwords.
Support role-based authorization.
Maintain audit logs.

Protect patient privacy.
Validate all user inputs.
Prevent SQL Injection.
Prevent Cross-Site Scripting (XSS).

7. Non-Functional Requirements

The system should:

Support at least 500 concurrent users.
Maintain 99.9% uptime.
Respond to API requests within 2 seconds.
Automatically back up data daily.
Be deployable using Docker.
Support PostgreSQL database.
Provide RESTful APIs.
Support future AI integrations.

8. Future Enhancements

Future versions may include:
AI-assisted diagnosis.
Medical image analysis.
Voice-based patient notes.
SMS appointment reminders.
Email notifications.
Video consultations.
Mobile applications (Android & iOS).
AI chatbot for patient assistance.

9. Acceptance Criteria

The project shall be considered complete when:
Patients can register and log in.
Doctors can manage appointments.
Medical records are stored securely.
Bills can be generated successfully.
Laboratory reports can be uploaded and downloaded.
Pharmacy inventory is managed correctly.
Role-based access control functions properly.
APIs are documented using OpenAPI/Swagger.